

Potential-Dependent Superlubricity of Ionic Liquids on a Graphite Surface

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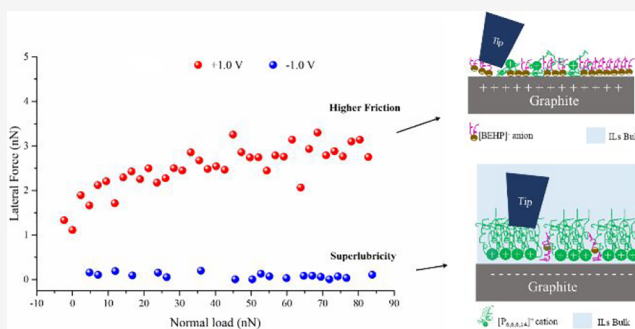
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ABSTRACT: The lubricities of four quaternary phosphonium ionic liquids—trihexyltetradecylphosphonium bis(2,4,4-trimethylpentyl)phosphinate ($[P_{6,6,6,14}][[C_8]_2PO_2]$), trihexyltetradecylphosphonium bis(2-ethylhexyl)phosphate ($[P_{6,6,6,14}][BEHP]$), trihexyltetradecylphosphonium bis(trifluoromethylsulfonyl)imide ($[P_{6,6,6,14}][TFSI]$), and tributylmethylphosphonium bis(trifluoromethylsulfonyl)imide ($[P_{4,4,4,1}][TFSI]$)—were measured as a function of potential on highly oriented pyrolytic graphite (HOPG) by using atomic force microscopy (AFM). The shear strength and surface contact radius values fitted from JKR model indicate the AFM probe slides on an IL boundary layer rather than bare HOPG. Frictions change as the compositions of the boundary layers switch from cation enriched to anion enriched when the potential changes from negative to positive. Superlubricity, which refers to near zero increase in friction with load, is achieved for the $[P_{6,6,6,14}]^+$ cation at -1.0 V and the $[TFSI]^-$ anion at $+1.0$ V. The lubricities of ILs are mainly influenced by three factors: the alkyl chain length, chemical composition, and ion sizes.



1. INTRODUCTION

Ionic liquids (ILs) are salts with melting points below 100 °C.^{1,2} ILs have attracted increasing research interests as lubricants and lubricant additives to reduce friction and wear for various solid surfaces³ due to physical properties including low vapor pressure, high chemical and thermal stability, and high thermal and electrical conductivity.^{4–7} In addition, the distinct interfacial nanostructures of ILs make them superior lubricant candidates, especially in the boundary lubrication regime, in which the applied normal load is high and the bulk lubricant liquid has been squeezed out.⁸

IL interfacial structure could be divided into three regimes: the bulk phase, near-surface layers, and the boundary (surface-adsorbed) layer.^{9,10} ILs including amphiphilic ions (usually the cation) often have sponge-like bulk nanostructures, where the charge centers of the ions aggregate together to form polar domains and neutral alkyl chains cluster to form apolar domains.¹¹ Near a solid surface, this sponge-like IL nanostructure is transformed to a flatter, more ordered, layered structure.¹² The boundary layer consists of ions in direct contact with the solid surface, and its composition depends on substrate properties including surface charge and roughness.¹³ When two surfaces separated by ILs are compressed, the near-surface layers are displaced at low load due to weak lateral cohesive forces.¹² The boundary layer is generally strongly bonded to the surface via strong electrostatic, van der Waals, hydrogen-bonding, and solvophobic

interactions and thus resists “squeeze out” up to normal loads of several gigapascals.¹⁴

For IL lubricants, boundary layer friction is controlled by the structure of ions and surface properties.^{15,16} Ion structures influence interionic interactions via electrostatic and van der Waals forces, as well as hydrogen bonding, and the resultant solvophobic interactions.¹⁵ Thus, different structured ions form boundary layers with variable roughness, robustness, and lubricity. For example, long alkyl chains tend to improve lubricity by increasing lateral solvophobic interactions which leads to a more robust boundary layer.^{17–19} Charge-localized ions are often more lubricating than charge-delocalized ions due to stronger electrostatic interactions with charged surfaces, which results in denser ion packing and more resistant boundary layers.²⁰ Fluorine, hydroxyl, carboxyl, and amide groups usually improve lubricity due to stronger interactions with surfaces via electrostatic and hydrogen-bonding interactions.^{20–22} The dimensions of ions can also affect the boundary layer lubrication,²³ for example, small spherical ions like iodide form neat, robust, lubricating boundary layers.¹⁹

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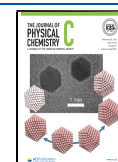
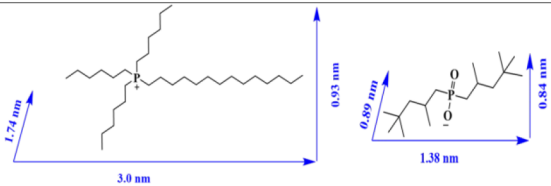
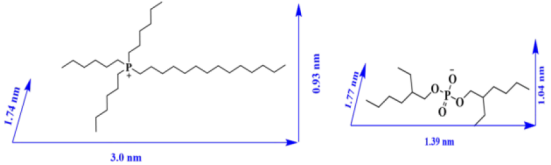
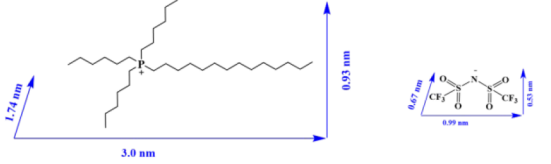
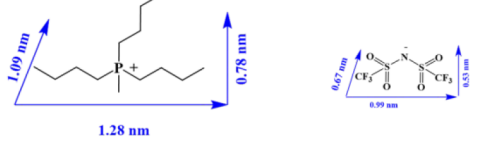


Table 1. Chemical Structures with 3D Dimensions of Ions Determined by Gaussian, Density ρ (25 °C), Molecular Weight M_w , and Ion Pair Dimension D_m Estimated by Assuming a Cubic Packing Geometry of the Selected ILs

ILs structure	ILs abbreviation	M_w (g·mol ⁻¹)	μ (g·cm ⁻³)	D_m (nm)
	[P _{6,6,6,14}] [(ⁱ C ₈) ₂ PO ₂]	805	0.89	1.14
	[P _{6,6,6,14}] [BEHP]	773	0.909	1.13
	[P _{6,6,6,14}] [TFSI]	763	1.065	1.06
	[P _{4,4,4,1}] [TFSI]	498	1.260	0.87

The energy dissipation pathways in boundary layer friction are generally proportional to the surface roughness of the sliding plane. There are two main pathways leading to energy dissipation while the AFM tip slides on the IL boundary layer: (1) compression, twist, rotation, and oscillation of the boundary layer ions and (2) expulsion of ion layers. The amount of energy dissipated via (1) is mainly determined by the ion structure, conformation, and packing density. The energy needed for process (2) is predominantly dependent on the lateral cohesive interactions between ions in the boundary layer and the innermost near surface layer as well minor contributions from subsequent near surface multilayers.²⁴

External potentials can be applied to conductive surfaces to tune the boundary layer composition and thereby control friction.^{25,26} For example, for Au(111) surfaces lubricated by ILs comprising the tris(pentafluoroethyl)trifluorophosphate anion ([FAP]⁻) with the 1-butyl-1-methylpyrrolidinium cation, the friction is low at negative potentials, and higher at positive potentials, because the cations form a smoother and more lubricating boundary layer.²⁷

Superlubricity is a motion regime in which both wear and friction nearly vanish, and the friction coefficient (μ) falls below 0.01.²⁸ Superlubricity has been detected between the sliding of 2D material layers due to the incommensurability of the rotating 2D material lattice.^{29–33} However, this type of superlubricity could be easily broken by rotation caused by

torque, thermal fluctuations, and high loads.^{29,34} High-pressure-induced superlubricity has been observed at amorphous carbon film surfaces³⁵ and in systems with carbon quantum dots^{36–38} as lubricant additives likely due to the *in situ* formation of graphene ribbons at the sliding interface under high contact pressure. However, using carbon quantum dots as lubricant additives usually leads to an initial “run-in” period, as carbon quantum dots adsorb to and react with the solid surface before the formation of a stable and smooth boundary film. Irreversible and inevitable surface wear occurs in this initial “run-in” period. Superlubricity has also been achieved by using ILs at ceramic interfaces due to the formation of a composite tribochemical layer; however, severe initial “run-in” periods are also found for these systems.^{39,40}

Our previous study has found that superlubricity could be “switched on” *in situ* at +1.5 V on HOPG likely due to the formation of a highly ordered anion enriched boundary layer for one conventional IL, 1-hexyl-3-methylimidazolium [FAP] ([HmIm][FAP]).⁴¹ This offers a convenient way to control superlubricity; however, as the nanofriction of only one IL has been investigated on HOPG as a function of potential, the effects of ion structures on superlubricity have not been probed. To address this knowledge gap, in this project, we have selected four phosphonium cation-based ILs and investigated how the lubricity varied with cation and anion structure, composition, and ion size at different potentials. Phosphonium

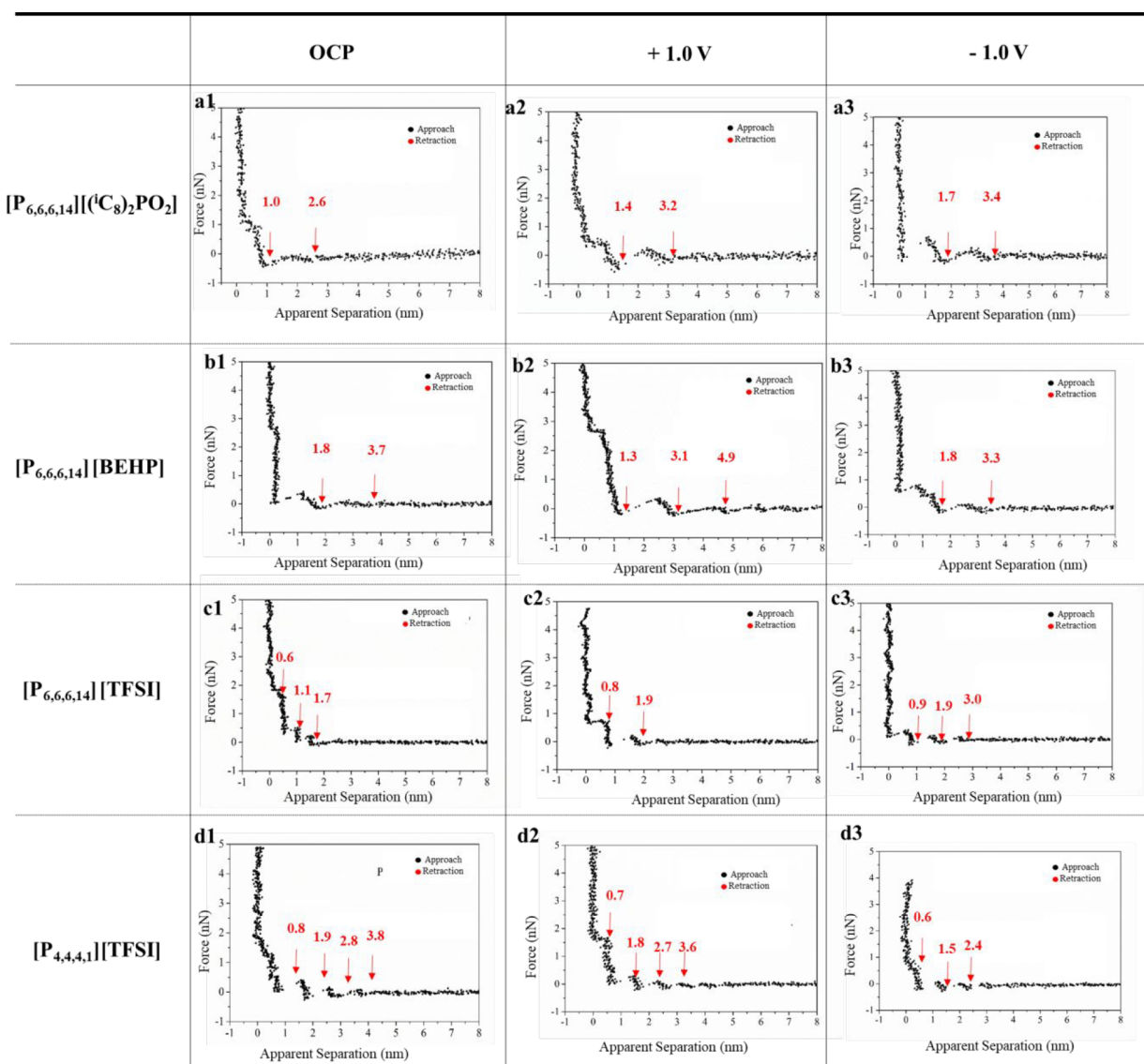


Figure 1. Typical normal force vs separation profile for (a) $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$, (b) $[P_{6,6,6,14}][\text{BEHP}]$, (c) $[P_{6,6,6,14}][\text{TFSI}]$, and (d) $[P_{4,4,4,1}][\text{TFSI}]$ on an HOPG surface. Panels a1–d1 are collected at OCP, a2–d2 at +1.0 V, and a3–d3 at –1.0 V. The starting positions of the steps are labeled in red.

cation-based ILs are employed in this work because they show better antifriction and antiwear performances than imidazolium cation-based ILs.²³ The outcomes from this work will assist the elucidation of potential-dependent lubricity mechanisms and the control of superlubricity, which can increase the energy efficiency and service life of machinery in the future

2. MATERIALS AND METHODS

The four ILs used in this study— $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$, $[P_{6,6,6,14}][\text{BEHP}]$, $[P_{6,6,6,14}][\text{TFSI}]$, and $[P_{4,4,4,1}][\text{TFSI}]$ —were provided by Iolitec Ionic Liquids Technologies GmbH (Germany) and used as received. The water contents, measured by Karl Fischer Titration (Mettler Toledo), were 0.19%, 0.16%, 0.11%, and 0.07%, respectively. The ion dimensions were determined by Gaussian⁴² and are shown with the physicochemical properties of the four ILs in Table 1.

HOPG (ZYB quality, TipsNano Co.) was cleaved freshly before each experiment. The surface potential was tuned from OCP (open circuit potential) to +1.0 V and then to –1.0 V by a WaveNano potentiostat for each experiment. AFM force

curve and friction measurements were performed by using a Veeco Nanoscope IV AFM in the contact mode. Sharp silicon AFM tips (NSC36, Mikromasch) with spring constants of $0.7 \pm 0.3 \text{ N}^{-1}$ and $\sim 8 \text{ nm}$ radius were cleaned with Milli-Q water and ethanol, followed by UV-ozone for 10 min before use. The AFM fluid cell (Bruker) was washed with Milli-Q water followed by ethanol and dried with nitrogen flow before experiment.

Normal force curves were collected by moving the HOPG surface toward the AFM tip and detecting the cantilever deflection as a function of separation. The ramp size and rate were 50 nm and 0.2 Hz, respectively. Standard methods were used to convert deflection vs separation data to normal force vs separation curves.⁴³ Lateral (frictional) forces were obtained by performing AFM scans with a scan angle of 90° (with respect to the long axis of the cantilever) and with the slow scan axis disabled.⁴³ The scan size was 500 nm, and scan rate was 6 Hz. The lateral deflection signal (i.e., cantilever twist) was converted to lateral force by using a customized function produced in MATLAB which takes into account the torsional

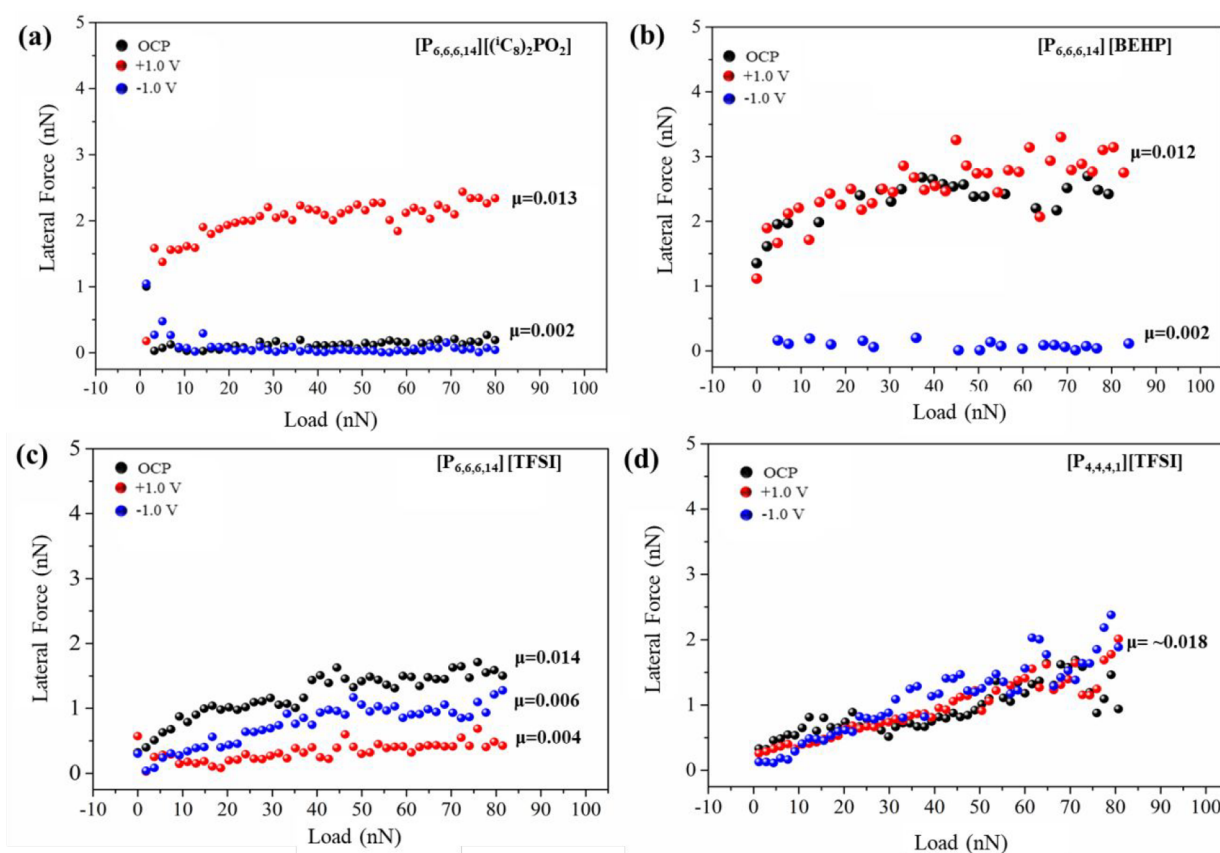


Figure 2. Lateral force vs normal load of four ILs at OCP, -1.0 V and $+1.0$ V on HOPG: (a) $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$, (b) $[P_{6,6,6,14}][\text{BEHP}]$, (c) $[P_{6,6,6,14}][\text{TFSI}]$, and (d) $[P_{4,4,4,1}][\text{TFSI}]$.

spring constant and the geometrical dimensions of the cantilever.⁴⁴ Every experiment was repeated at least three times. Experimental results show friction is reversible under potential within 10 min (Figure S1).

3. RESULTS AND DISCUSSION

3.1. Interfacial Nanostructure. Figure 1 shows the normal force curves of four ILs on the HOPG surface. As the AFM tip approaches the surface, discrete steps are detected, similar to other ILs.^{45,46} For $[P_{6,6,6,14}][\text{TFSI}]$ and $[P_{4,4,4,1}][\text{TFSI}]$ (Figures 1c and 1d), the step widths are ~ 1.1 and ~ 0.9 nm, respectively, which agree with the calculated ion pair dimensions assuming cubic packing (Table 1), in accordance with prior results for other ILs.^{19,46,47} However, the step widths of both $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$ and $[P_{6,6,6,14}][\text{BEHP}]$ are ~ 1.8 nm (Figures 1a and 1b), which is significantly larger than the calculated ion pair dimensions of 1.1 nm (Table 1). This is likely caused by steric effects of branched alkyl tails in $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$ and $[P_{6,6,6,14}][\text{BEHP}]$, which invalidates the cubic packing approximation. Radiation scattering experiments are planned to resolve the liquid nanostructure of these ILs and will be reported in the future.

The step height in the normal force curve corresponds to the force required to push through the near surface layer and is determined by the cohesive forces between ions in the film. Previous studies have shown that even very stiff AFM cantilevers at maximum force could not push through the strongly adsorbed ion boundary layer at the interface.^{14,48–50} Thus, here the final push-through force is assumed to be the

minimum normal force (applied on the AFM probe) required to reach the boundary layer. The final push-through forces for all four ILs are below 4 nN (cf. Figure 1), which means the majority of the friction data in Figure 2 were collected in the boundary layer friction regime.

3.2. Friction. Figure 2 shows the lateral (frictional) forces vs normal loads of the four ILs as a function of potential. In all cases, friction is lower than that measured in air, indicating these ILs are good lubricants for HOPG surfaces.⁴¹

The variation of friction with applied potential is different for different ILs (cf. Figure 2). For example, $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$ shows almost vanishing friction at -1.0 V and OCP, but much higher friction at $+1.0$ V, whereas $[P_{6,6,6,14}][\text{TFSI}]$ shows the highest friction at OCP, followed by -1.0 V, and the lowest friction at $+1.0$ V. Furthermore, superlubricity ($\mu < 0.01$) is detected in a few cases, including -1.0 V and OCP for $[P_{6,6,6,14}][(\text{C}_8)_2\text{PO}_2]$, -1.0 V for $[P_{6,6,6,14}][\text{BEHP}]$, and both $+1.0$ and -1.0 V for $[P_{6,6,6,14}][\text{TFSI}]$.

Applying a potential on HOPG can change the composition of the ion layer adsorbed to the solid surface,⁵¹ as well as the conformation and orientation of ions, and thus alter the lubricity of the boundary layer.^{41,52} At OCP, boundary layers are a mixture of cations and anions, with the cation:anion ratio determined by the relative affinity of each ion for the surface. On the basis of previous work, it is reasonable to assume that, in the absence of potential, solvophobic interactions between the surface and the ions drive adsorption, and the adsorbed ions tend to lie along the HOPG surface in a horizontal monolayer to maximize alkyl chain–surface attractions.^{51–53} At -1.0 V, the boundary layer is cation enriched. The cationic

Table 2. Shear Strength S_c and the Surface Contact Radius (a) at a Load of 20 nN from JKR Fits

	[P _{6,6,6,14}][(ⁱ C ₈) ₂ PO ₂]			[P _{6,6,6,14}][BEHP]			[P _{6,6,6,14}][TFSI]			[P _{4,4,4,1}][TFSI]		
	OCP	+1 V	−1 V	OCP	+1 V	−1 V	OCP	+1 V	−1 V	OCP	+1 V	−1 V
$S_{c(\text{JKR})}$ (MPa)	4.2	5.5	0.2	0.2	4.4	0.05	2.1	0.4	1.0	1.5	1.3	1.5
$a(F_N = 20 \text{ nN, nm})$	14	13	16	14	12	21	11	13	12	11	11	11

charge centers tend to associate with the negatively charged surface, and the neutral alkyl tails point toward the bulk and interact laterally via van der Waals forces. At +1.0 V the boundary layer is anion enriched with charge centers pointing toward the positively charged surface. The following discussion explains the friction behavior in the boundary regime, with a normal load greater than 4 nN, in detail and illustrates mechanisms to achieve superlubricity.

According to the modified Amontons' law,⁵⁴ the total lateral force (F_{LT}) required for one surface to move across another could be expressed by

$$F_{LT} = S_c \pi a^2 + \mu F_N \quad (1)$$

where S_c is the shear strength, a is the surface contact radius, μ is the friction coefficient, and F_N is the load. The first term is controlled by the adhesion; its magnitude is governed by both surface roughness and the physiochemical interactions between surfaces. The second term is controlled by the load; its magnitude is determined predominately by surface topography and roughness.

For a rigid surface, the surface contact radius (a) is usually small and almost constant as the load increases; thus, the lateral force is mainly controlled by the load.⁵⁴ However, as shown in Figure 2, the lateral force does not increase with the load linearly for most cases, especially when the load is lower than 20 nN. The curved shapes of the lateral force vs load profiles are similar to those detected on graphene monolayer,⁵⁵ indicating that the total lateral force is affected by the adhesion at least when the load is low for these systems. S_c and a in the adhesion-controlled term could be coarsely estimated by using different contact mechanics models; however, the Johnson–Kendall–Roberts (JKR) model has been found to fit the lateral force data better than other models for graphene systems.⁵⁵ The solid surface used in this study is HOPG, which can be considered as multilayer graphene at the nanoscale; thus, the JKR model is also selected to fit the lateral force data in the lower load regime ($F_N < 20 \text{ nN}$) in this study. The details on how to achieve S_c and a are described in the Supporting Information, and the JKR fitted values of S_c and a at a load of 20 nN are listed in Table 2.

The shear strength (S_c) is mainly determined by the interactions between the two contact surfaces such as H-bonding.⁵⁴ The JKR fitted S_c values for all ILs at all potentials are below 6 MPa, which are more than 1 order of magnitude lower than the values obtained previously^{55,56} for bulk HOPG and monolayer graphene. This strongly supports our argument that the AFM probe slides on top of an IL boundary layer instead of the bare HOPG surface during the friction measurement. Furthermore, Table 2 shows that S_c values for [ⁱC₈)₂PO₂][−] and [BEHP][−] enriched boundary layers are higher than the [TFSI][−] anion enriched layer at +1.0 V and cation enriched boundary layers at −1.0 V as well as mixed boundary layers at OCP. This suggests that the boundary layers enriched in [ⁱC₈)₂PO₂][−] and [BEHP][−] could form more H-bonds with the silicon AFM probe (always covered

with a native silicon oxide layer)⁵⁷ than other ions, likely via O=P=O with Si–OH.

The surface contact radii (a) at a load of 20 nN are obviously higher for [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−] than [P_{6,6,6,14}][TFSI][−] at −1.0 V (Table 2). As [P_{6,6,6,14}]⁺ cations have much longer and more compressive alkyl chains than the anions, these results strongly indicate that although the boundary layers of all these ILs are enriched in [P_{6,6,6,14}]⁺ cations, the cation ratios for [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−] are higher than [P_{6,6,6,14}][TFSI][−], leading to more compressive boundary layers and larger surface contact radii. At +1.0 V, for all three [P_{6,6,6,14}]⁺ ILs, the [TFSI][−] enriched boundary layer is expected to be less compressive than the [ⁱC₈)₂PO₂][−] and [BEHP][−] enriched boundary layer since [TFSI][−] has much shorter and less branched chains. However, a is similar for these three ILs. This result indicates the coexistence of cations in the anion enriched boundary layer, and the cation ratio is higher for [P_{6,6,6,14}][TFSI][−] than for [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−]. At OCP, a is also slightly higher for [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−] than [P_{6,6,6,14}][TFSI][−], which means the boundary layer composed of mixed ions is more compressive for the former two ILs than the last. The surface contact radii of [P_{4,4,4,1}][TFSI][−] are smaller than the other three ILs at all potentials, which are in line with the shorter chains and less compressibility of both [P_{4,4,4,1}]⁺ cation and [TFSI][−] anion.

When the load is higher than 20 nN, the lateral force increases slowly but linearly with the load and does not follow the JKR model for in most cases, as shown in Figure 2 and Figure S2. This means the boundary layers and subsequent HOPG surfaces are fully compressed, and the surface contact radii do not increase significantly; the lateral forces are mainly load-controlled. The friction coefficients (μ) extracted from the gradient of the lateral force vs normal load in these linear regions (listed in Figure 2) are discussed in the following section to elucidate how friction changes with boundary layer composition and structure in this load-controlled regime.

Both [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−] present nearly vanishing friction ($\mu = 0.002$) at −1.0 V as the load increases higher than 20 nN, indicating that [P_{6,6,6,14}]⁺ enriched boundary layers are superlubricating (cf. Figure 2a,b). This suggests even at full compression, the long and straight [P_{6,6,6,14}]⁺ alkyl chains have strong lateral cohesive interactions, which lead to the formation of smooth and robust boundary layers to resist “squeeze out” during sliding, similar to results for other ILs⁵⁸ and self-assembled alkane monolayers.^{17,59,60} To the best of our knowledge, it is the first time that superlubricity has been measured on negatively charged HOPG surfaces. At −1.0 V, the friction coefficient of [P_{6,6,6,14}][TFSI][−] is slightly higher than [P_{6,6,6,14}][(ⁱC₈)₂PO₂][−] and [P_{6,6,6,14}][BEHP][−]. This indicates that although enriched in [P_{6,6,6,14}]⁺ cations, there are sufficient anions present in the boundary layer, leading to different lubricities, consistent with the surface contact radius results discussed previously. The ion size of [TFSI][−] is much smaller than [ⁱC₈)₂PO₂][−], [BEHP][−], and [P_{6,6,6,14}]⁺ (as shown in Table 1); thus, the boundary layer

of $[P_{6,6,6,14}][TFSI]$ composed of mismatched ions is more loosely packed and rougher than either $[P_{6,6,6,14}][(^iC_8)_2PO_2]$ or $[P_{6,6,6,14}][BEHP]$, where the ions are similarly sized, resulting in higher friction. At -1.0 V, the $[P_{4,4,4,1}]^+$ cation enriched boundary layer shows higher friction coefficients than $[P_{6,6,6,14}]^+$ cation (cf. Figures 2c and 2d). This is consistent with the previous work⁵⁸ and is consistent with shorter alkyl chains having weaker lateral interactions, which leads to less robust boundary layers and reduced lubricity.

At $+1.0$ V, for both $[P_{6,6,6,14}][(^iC_8)_2PO_2]$ and $[P_{6,6,6,14}][BEHP]$, the friction coefficients are much higher than at -1.0 V, indicating the anion enriched boundary layer is less lubricating than the cation enriched boundary layer. Compared with the $[P_{6,6,6,14}]^+$ cation, both $[(^iC_8)_2PO_2]^-$ and $[BEHP]^-$ anions have branched alkyl chains, which hinder anion packing in the adsorbed layer and thereby reduce lateral interactions.²⁴ Consequently, $[(^iC_8)_2PO_2]^-$ and $[BEHP]^-$ enriched boundary layers are less robust than $[P_{6,6,6,14}]^+$ enriched layers, resulting in more energy dissipation and higher friction at $+1.0$ V. For $[P_{6,6,6,14}][TFSI]$, the $[TFSI]^-$ enriched boundary layer is superlubricating at $+1.0$ V, in accordance with previous results for different fluorine-containing anions.⁴¹ This is likely because the dense packing of bulky fluorine atoms reduces geometric contact corrugation, which minimizes steric barriers for sliding⁶¹ and lowers friction. The friction coefficient of $[P_{4,4,4,1}][TFSI]$ is higher than that of $[P_{6,6,6,14}][TFSI]$ at $+1.0$ V (cf. Figures 2c and 2d). This confirms the presence of cations in the boundary layer at positive potentials, and cations with longer alkyl chains experience stronger lateral interactions which lead to a better formed boundary layer and lower friction.

At OCP, the friction of $[P_{6,6,6,14}][(^iC_8)_2PO_2]$ is close to zero, with data similar to that obtained at -1.0 V (cf. Figure 2a). However, the friction of $[P_{6,6,6,14}][BEHP]$ at OCP is much higher and closer to the friction measured at $+1.0$ V for this IL (Figure 2b). These results are likely caused by the composition and conformation of ions in boundary layers. For $[P_{6,6,6,14}][(^iC_8)_2PO_2]$, the quaternary carbon groups of $[(^iC_8)_2PO_2]^-$ anions are sterically hindered, which impedes them from lying parallel along the surface, which results in the boundary layer at OCP being enriched in $[P_{6,6,6,14}]^+$ cations, similar to -1.0 V. In contrast, for $[P_{6,6,6,14}][BEHP]$ anions preferentially adsorb to HOPG at OCP because $[BEHP]^-$ anions can more readily adopt a flat orientation on the surface due to the two short $-O^-$ groups (compared to the C_6 groups attached to the phosphonium cation); thus, the friction is similar to that measured at $+1.0$ V, where the boundary layer is anion enriched. For $[P_{6,6,6,14}][TFSI]$, the friction at OCP is higher than that at both $+1.0$ and -1.0 V. The ion sizes of $[P_{6,6,6,14}]^+$ cations and $[TFSI]^-$ anions are very different (Table 1), making the mixed boundary layer at OCP rough and loosely packed, leading to friction. When replacing the large $[P_{6,6,6,14}]^+$ cation with a smaller $[P_{4,4,4,1}]^+$ cation, the friction at OCP is similar to that at both positive and negative potentials. This confirms that the matching of ionic dimensions is a significant factor to control boundary layer friction, especially at OCP.

4. CONCLUSION

The frictions of four phosphonium cation-based ILs— $[P_{6,6,6,14}][(^iC_8)_2PO_2]$, $[P_{6,6,6,14}][BEHP]$, $[P_{6,6,6,14}][TFSI]$, and $[P_{4,4,4,1}][TFSI]$ —have been found to change as a function of potential on the HOPG surface. Cations have been found to be present in anion enriched boundary layers, contributing to

lubricity at positively charged surfaces, and vice versa. The lubricity has been found to be affected by three factors, i.e., alkyl chain length, chemical composition, and ion size. Superlubricity can be switched on in the following cases: (i) $[P_{6,6,6,14}]^+$ enriched boundary layer at -1.0 V, where long alkyl chains associated with $[P_{6,6,6,14}]^+$ cations assist the formation of robust boundary layers via enhancing lateral cohesive forces, and (ii) $[TFSI]^-$ enriched boundary layer at $+1.0$ V, where the bulky fluorine atoms reduce geometric contact corrugations and form a smooth and flat boundary layer.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcc.0c10804>.

Repeated data of $[P_{6,6,6,14}][(^iC_8)_2PO_2]$ on HOPG; Johnson–Kendall–Roberts (JKR) model fitting (PDF)

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